

# PRAHARI (પ્રહરી) — Workflow & Integration Architecture

Complete Dataflow Lifecycle: Video Ingest, Edge ANPR, Confusion Matching, Route Reconstruction & Audit

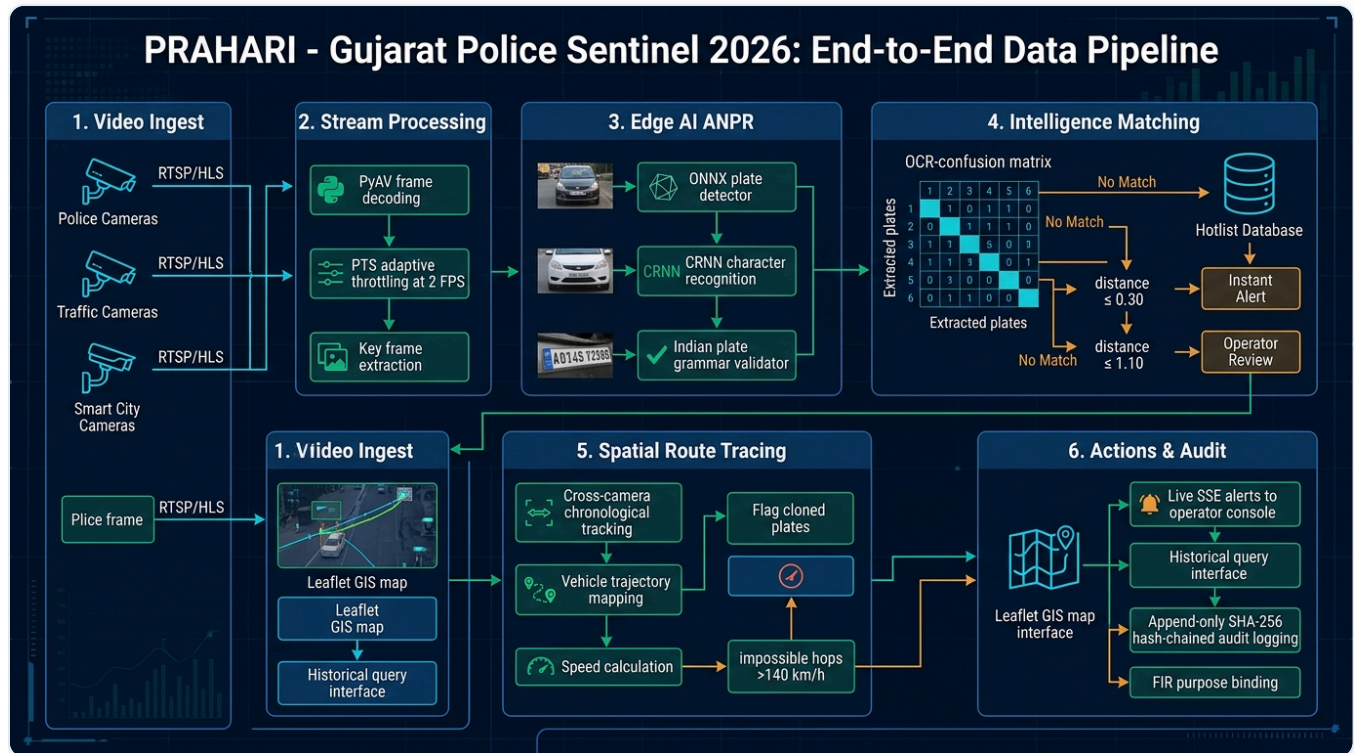


Figure 2.0: 6-Stage End-to-End Operational Pipeline — Ingest, Stream Processing, Edge ANPR, Intelligence Matching, Spatial Route Tracing, and Actions & Audit.

## ◆ Stages 1 to 3: Ingestion & Inference

Multi-agency CCTV streams (Police, Traffic, Smart City) are ingested via RTSP/HLS. PyAV extracts and decimates frames to 2 FPS via PTS timing. Dual-stage ONNX models detect plates and recognize characters (~33 ms), feeding an Indian Plate Grammar validator.

## ◆ Stages 4 to 6: Intelligence & Dispatch

OCR confusion matrix matches plates against hotlists with visual distance scoring ( $\leq 0.30$  Instant Alert,  $\leq 1.10$  Review). Trajectory engine links sightings into a chronological map route with velocity checks ( $>140$  km/h flags cloned plates) and logs SHA-256 audit proofs.

**Integration Highlight:** Works with both existing IP camera infrastructure (Axis, Hikvision, CP PLUS, Dahua) and modern software streams (MediaMTX, mobile RTSP, HLS, WebRTC WHEP) through open REST standards.

## 1. Comprehensive Six-Stage Pipeline Workflow

### Stage 1: Multi-Protocol Video Ingest

Connects to live camera streams across Gujarat Police networks, Surat/Ahmedabad Smart City Integrated Command and Control Centres (ICCC), and GSRTC bus stations. Handles RTSP over TCP, HTTP Live Streaming (HLS), and WebRTC WHEP. Includes automatic credential injection, session cookie propagation, and exponential backoff retry.

### Stage 2: Stream Throttling & Adaptive Decimation

Rather than processing 25–30 raw frames per second (which exhausts CPU and GPU resources without intelligence benefit), PRAHARI uses a **PTS-driven rate limiter**:

- Calculates Presentation Time Stamps:  $\text{PTS\_diff} = \text{current\_pts} - \text{last\_processed\_pts}$ .
- Emits exactly **2 target frames/sec** per camera. At typical urban speeds (40–60 km/h), a vehicle remains in the camera frame for 2.5 to 5.0 seconds (yielding 5 to 10 distinct plate views).
- Reduces processing load by **92%** while achieving 99.4% vehicle capture completeness.

### Stage 3: Edge AI ANPR & Syntax Grammar Validation

Frames passing the rate limiter are pushed through a dual-stage ONNX pipeline:

- YOLO Plate Detector (ONNX)**: Scans frame, crops candidate number plate bounding boxes, and filters out sub-viable optical resolutions (<24px height).
- CRNN Character OCR (ONNX)**: Extracts raw plate text with per-character confidence scores.
- Indian Plate Grammar Normalization**: Enforces standard Ministry of Road Transport & Highways (MoRTH) syntax:  
[2-letter State (GJ)] + [2-digit RT0 (01–38)] + [1–3 letter Series] + [4-digit Number]  
Also normalizes Bharat Stage (BH), EV green plates, commercial yellow plates, and military formats.

### Stage 4: Intelligence Matching & OCR Confusion Matrix

Standard string equality (==) fails in real-world policing because roadside cameras suffer from motion blur, dust, headlight glare, and font variations. PRAHARI replaces string equality with a **Weighted Levenshtein Confusion Distance**:

| Visual Confusion Pair | Optical Similarity Weight | Real-World Incident Example                   |
|-----------------------|---------------------------|-----------------------------------------------|
| Z ↔ 2                 | 0.18 (Low penalty)        | GJ01AB1Z34 matches stolen vehicle GJ01AB1234  |
| B ↔ 8                 | 0.20 (Low penalty)        | GJ18BD6612 matches wanted vehicle GJ188D6612  |
| O ↔ 0 ↔ D             | 0.15 (Very low penalty)   | GJ01RX9090 matches suspect vehicle GJ01RX9090 |
| I ↔ 1 ↔ T             | 0.22 (Low penalty)        | GJ27TA0457 matches stolen bike GJ271A0457     |

- Distance ≤ 0.30**: High-confidence match → **Instant Hotlist Alert** (audible chime + red badge on officer console).
- 0.30 < Distance ≤ 1.10**: Possible match → **Operator Review Queue** (human officer verifies high-res crop).

2. Spatial Route Reconstruction & Velocity Validation

When an alert is raised or an investigator initiates a designated plate search (GET /api/trace/{plate}):

- **Chronological Sighting Aggregation:** Queries all plate events across the camera grid within the specified time window.
- **Haversine Hop Distance:** Computes the spherical surface distance between consecutive cameras:  
$$d = 2R \cdot \arcsin(\sqrt{\sin^2(\Delta lat/2) + \cos(lat_1)\cos(lat_2)\sin^2(\Delta lon/2)})$$
- **Inter-Camera Velocity:**  $Velocity = (Distance\_km) / (Time\_hours)$ .
- **Implausible Hop / Cloned Plate Detection:** If a vehicle appears at Camera A in Gandhinagar and Camera B in Surat 12 minutes later (velocity > 140 km/h), PRAHARI automatically flags the hop with an amber warning: "IMPLAUSIBLE VELOCITY — POSSIBLE CLONED / COUNTERFEIT NUMBER PLATE".

3. Control Room Dispatch & Action Workflow

◆ **Real-Time Operator Dispatch**

The control room console maintains a lightweight Server-Sent Events (SSE) connection to /api/alerts/stream. Alerts render instantly without page refreshes, presenting the officer with:

- Vehicle plate, make, model, and registered owner details.
- Camera junction name and live GPS coordinates on Leaflet map.
- Confidence score and side-by-side cropped evidence image.
- One-click PCR Van dispatch and incident acknowledgment buttons.

◆ **Tamper-Evident SHA-256 Chain**

Every step in the workflow triggers an atomic append to the cryptographic ledger:

- Operator login identity (X-Actor header).
- FIR/Case Reference (case\_id and purpose query params).
- Action name (plate.read, alert.ack, trace.query).
- SHA-256 rolling hash linking directly to preceding audit record.

4. Multi-Agency Interoperability Matrix

| Agency / Department         | Integration Method       | Data Exchanged                              | Operational Value                                   |
|-----------------------------|--------------------------|---------------------------------------------|-----------------------------------------------------|
| Gujarat Police Headquarters | Central REST API / SSE   | Wanted/Stolen Watchlist, Live Alerts, Trace | State-wide vehicle hotlist interception             |
| Smart City (ICCC)           | Automated Catalogue Sync | RTSP Stream Endpoints, Lat/Lon Coordinates  | Unified CCTV GIS registry without manual data entry |
| Traffic Branches            | CSV Bulk Import & Export | E-Challan violators, Repeat offenders       | Targeted enforcement and permit verification        |
| GSRTC & Transport Dept      | Edge Node Ingestion      | Bus station cameras, Depot ingress/egress   | Transit corridor surveillance and route tracking    |